

ϕ TD-Gaussian

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.06141	0.14	0.1202	0.0842
Cramér $\ell_2^2 \downarrow$	0.0006766	0.002284	0.002927	0.001618
CF - $\ell_2^2/\omega^2 \downarrow$	0.0005699	0.001558	0.00256	0.001342

ϕ TD-Cauchy

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.1197	0.03785	0.07609	0.1322
Cramér $\ell_2^2 \downarrow$	0.003941	0.0001825	0.001764	0.002499
CF - $\ell_2^2/\omega^2 \downarrow$	0.002608	0.0001576	0.0008532	0.001347

ϕ TD-Gamma

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.7164	2.58	1.955	0.1258
Cramér $\ell_2^2 \downarrow$	0.2984	0.8119	0.7406	0.001809
CF - $\ell_2^2/\omega^2 \downarrow$	0.2627	0.7464	0.7005	0.001861

ϕ TD-Categorical

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.2832	0.3909	0.3348	0.309
Cramér $\ell_2^2 \downarrow$	0.0281	0.0187	0.03766	0.02862
CF - $\ell_2^2/\omega^2 \downarrow$	0.02106	0.01475	0.03094	0.02297

ϕ TD-Quantile

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.1011	0.2179	0.1353	0.1815
Cramér $\ell_2^2 \downarrow$	0.002147	0.003016	0.003938	0.00369
CF - $\ell_2^2/\omega^2 \downarrow$	0.001598	0.003516	0.003357	0.003258